

Chapter 5

Software Scheduling:

- **Project planning Steps:**

- 1) **Establish project scope:**

Understand the problem and the work that must be done.

- 2) **Determine feasibility:**

To determine whether it would be financially and technically feasible to develop the product.

- 3) **Analyze risks:**

What can go wrong? How can we avoid it? What can we do about it?

- 4) **Define required resources**

- a) Determine human resources required
- b) Define reusable software resources
- c) Identify environmental resources

- 5) **Estimate cost and effort**

- a) Decompose the problem
- b) Develop two or more estimates using different approaches

- 6) **Develop a project schedule**

- a) Establish a meaningful task set
- b) Define a task network
- c) Use scheduling tools to develop a timeline chart
- d) Define schedule tracking mechanism

Risk Analysis and Management:

Software risks:

- What can go wrong?
- What is the likelihood?
- What will be the damage?
- What can be done about it?

Risk analysis and management are a set of activities that help a software team to understand and manage uncertainty about a project.

Risk Management

The process by which a course of action is selected that balances the potential impact of a risk weighted by its probability of occurrence and the benefits of avoiding (or controlling) the risk

Risk management life cycle:

- a. Identify(risk identification)
- b. Analyze(risk analysis)
- c. Plan(contingency planning)
- d. Track(risk monitoring)
- e. Control(recovery management)



- **Software Project Scheduling:**

Software project scheduling distributes estimated effort across the planned project duration by allocating the effort to specific tasks.

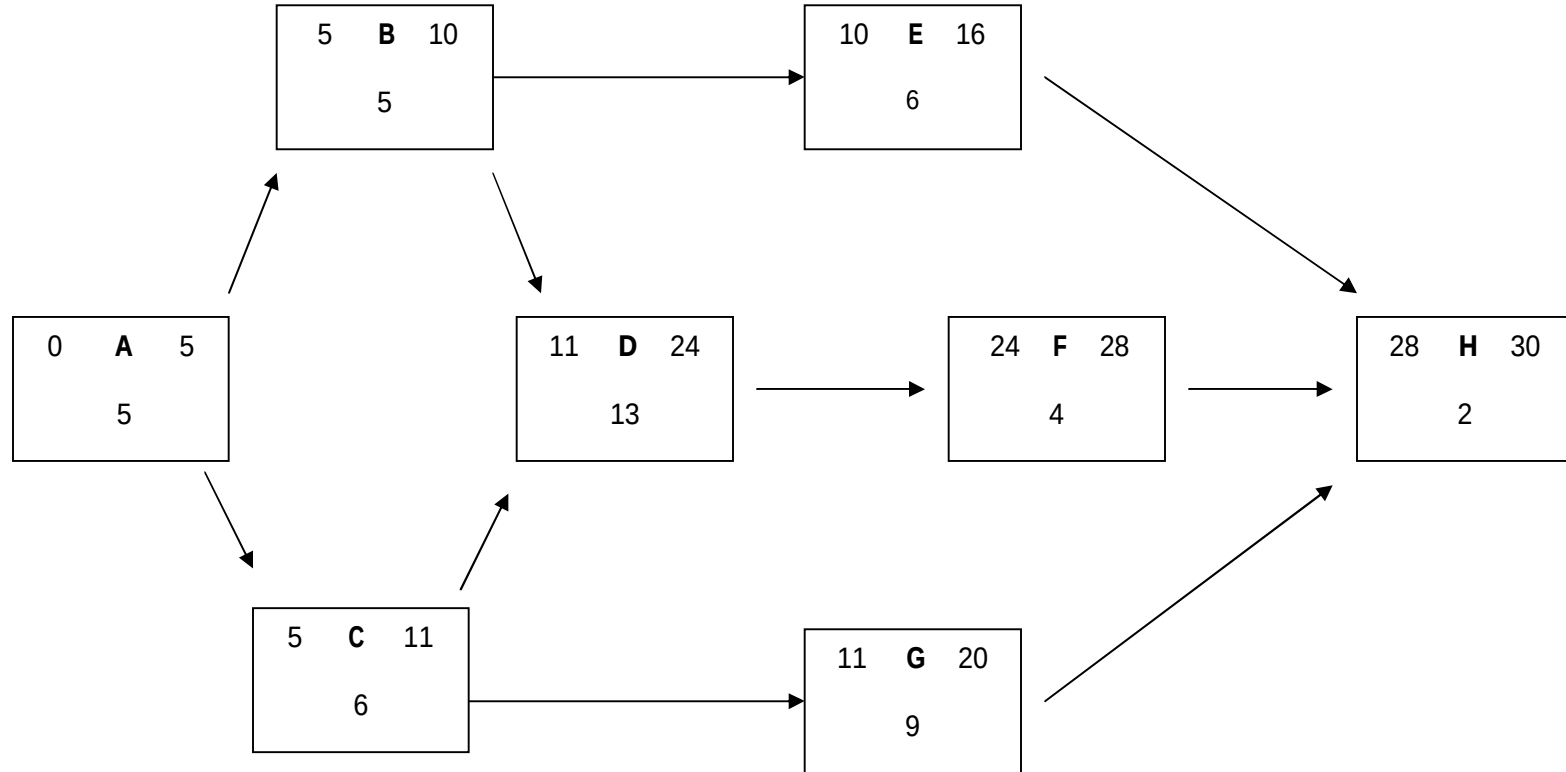
Basic principles of project scheduling:

- **Compartmentalization**
 - The project must be compartmentalized into a number of manageable activities, actions, and tasks.
- **Interdependency**
 - The interdependency of each compartmentalized activity, action, or task must be determined
 - Some tasks must occur in sequence while others can occur in parallel
 - Some actions or activities cannot commence until the work product produced by another is available
- **Time allocation**
 - Each task to be scheduled must be allocated some number of work units
 - In addition, each task must be assigned a start date and a completion date that are a function of the interdependencies.
- **Effort validation**
 - Every project has a defined number of people on the team
- **Defined responsibilities**
 - Every task that is scheduled should be assigned to a specific team member
- **Defined outcomes**
 - Every task that is scheduled should have a defined outcome for software projects such as a work product or part of a work product
- **Defined milestones**
 - A milestone is accomplished when one or more work products has been reviewed for quality and has been approved

<u>Activity</u>	<u>Predecessors</u>	<u>Estimated Duration</u>
A	None	5
B	A	5
C	A	6
D	B, C	13
E	B	6
F	D	4
G	C	9
H	E, F, G	2

1. Subtract activity times along each path as you move through the network ($LF - \text{Dur} = LS$),
2. Carry back the LS time to the activity nodes immediately preceding the successor node. That LS becomes the LF of the next node, unless the preceding node is a burst point.
3. In the case of a burst point, the smallest succeeding LS become the LF for that node.

ES(Earliest Start)	Task Name	Earliest Finish(EF)
Duration		



Critical path analysis:

A project-management technique that lays out all the activities needed to complete a task, the time it will take to complete each activity and the relationships between the activities. A task on the Critical Path has no (zero) slack.

Possible paths: A-B-E-H=5+5+6+2=18

A-B-D-F-H = 5+5+13+4+2=29

A-C-D-F-H = 5+6+13+4+2=30 /critical path

$$A-C-G-H = 5+6+9+2=22$$

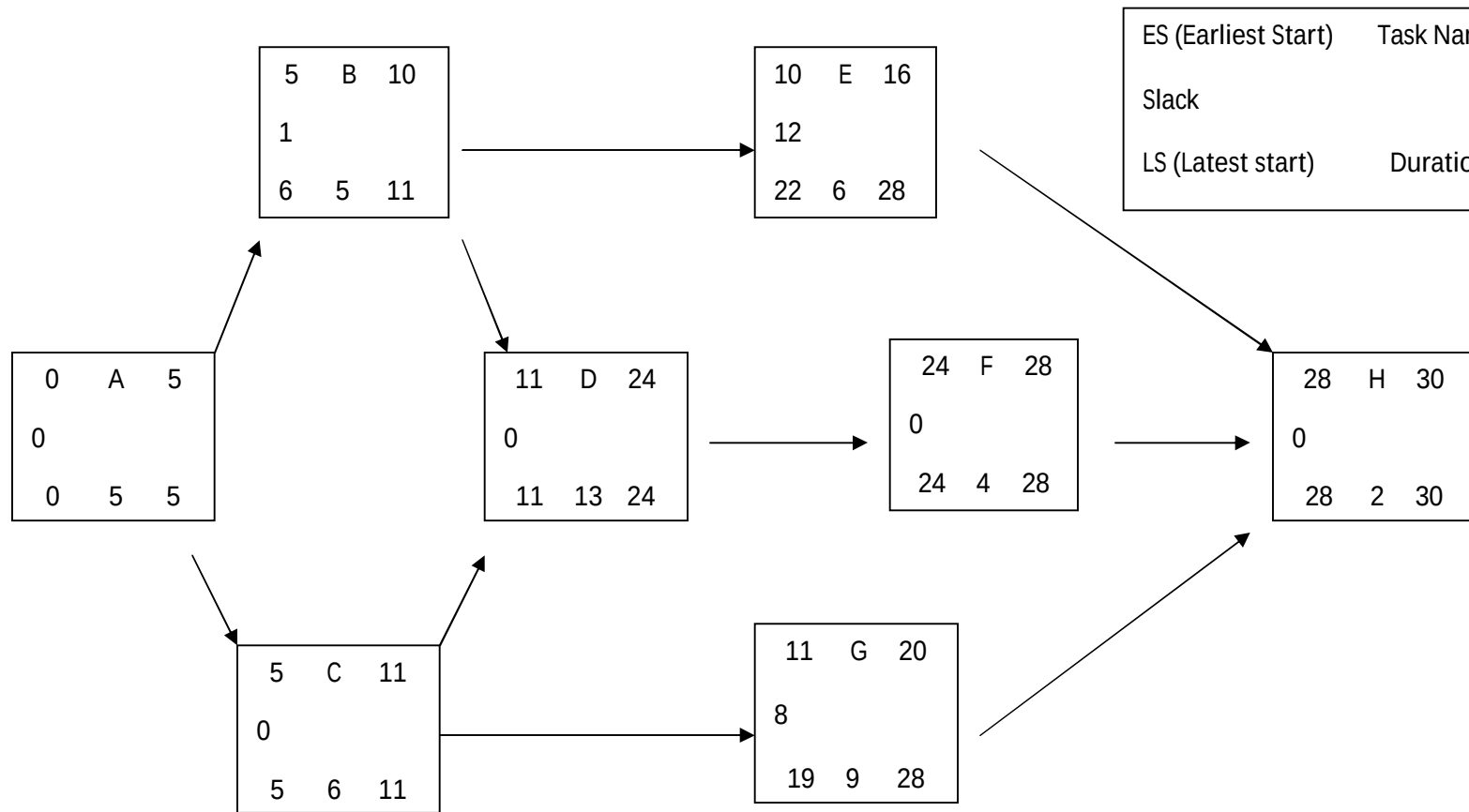
2. How is slack determined?

Slack for a particular task is calculated by:

- Subtracting the earliest time the task can start from the latest time the task can start (LS - ES).

3. How do you determine the ES for an activity with two predecessors? How do you determine the LF for an activity with two successors?

- The **ES** for an activity with two predecessors is equal to the **later EF** of the two predecessors since both predecessors must be completed for the task to begin.
- The LF for an activity with two successors is equal to the **smaller** LS of the two successors.



ES (Earliest Start)	Task Name	EF (earliest finish)
Slack		
LS (Latest start)	Duration	LF (Latest Finish)

